

Quiz Result

Adavance DSA



Score

12.399999999999999 / 26



Percentage

47.69%



Category

Programming

Question 1

Incorrect

 Question

What are the applications of a Segment Tree?

 Your Answer

used for operting System

Correct Answer Score : 0 / 1Evaluation Method : **LLM** **Feedback**

The answer is completely irrelevant to the question about Segment Trees. No marks awarded.

Explanation

The context mentions that Segment Trees are used for range queries, which can be expanded to describe their applications.

Question 2**Correct** **Question**

Which of the following is NOT a property of a Red Black Tree?

 Your Answer

Two red nodes can be adjacent

Correct Answer

Two red nodes can be adjacent

 Score : 1 / 1Evaluation Method : **NODE****Explanation**

The context states that no two red nodes can be together in a Red Black Tree.

Question 3

Incorrect

Question

Explain the collision handling techniques in a Hash Table.

Your Answer

Linear Probaling

Correct Answer

 Score : 0.2 / 1

Evaluation Method : [LLM](#)

Feedback

The answer is incomplete and contains a spelling error ('probaling' instead of 'probing'). It only mentions one minor part of open addressing and does not cover the main techniques (chaining and open addressing) or their methods.

Explanation

The context mentions Chaining and Open Addressing as collision handling techniques in a Hash Table.

Question 4

Correct

Question

Describe the applications of a Heap.

Your Answer

It is used to memory manage

Correct Answer

 Score : 1 / 1

Evaluation Method : [Embedding](#)

 Feedback

Answer is correct

Explanation

The context specifies that Heaps are used for Priority Queues and Heap Sort.

Question 5

Correct

 Question

What are the key properties of a B Tree?

 Your Answer

The context does not provide specific properties of a B Tree, but it mentions that B Trees are used in database indexing and file systems. Typically, B Trees are balanced tree structures that allow efficient insertion, deletion, and search operations, and they can have multiple keys per node.

Correct Answer

 Score : 1 / 1

Evaluation Method : [Embedding](#)

 Feedback

Answer is correct

Explanation

While the context does not detail B Tree properties, it mentions their applications, which can be used to infer their importance.

Question 6

Incorrect

Question

Which of the following is NOT an application of BFS?

Your Answer

All are applications

Correct Answer

Cycle Detection

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

The context lists Shortest Path and Networking as applications of BFS, while Cycle Detection is an application of DFS.

Question 7

Correct

Question

How does a Trie improve the efficiency of string search operations?

Your Answer

"A Trie improves the efficiency of string search operations by storing strings in a tree-like structure where each node represents a character. This allows for $O(\text{length of word})$ search

time, as each character in the word is checked sequentially. Tries are particularly efficient for prefix-based searches and autocomplete features."

Correct Answer

 Score : 1 / 1

Evaluation Method : [Embedding](#)

Feedback

Answer is correct

Explanation

The context mentions that Tries are used for storing strings and that search operations take $O(\text{length of word})$ time.

Question 8

Correct

Question

Which of the following is NOT a type of graph mentioned in the context?

Your Answer

Cyclic Graph

Correct Answer

Cyclic Graph

 Score : 1 / 1

Evaluation Method : [NODE](#)

Explanation

The context lists Directed, Undirected, and Weighted Graphs but not Cyclic Graph.

Question 9

Correct

Question

Explain the difference between a Max Heap and a Min Heap.

Your Answer

In a Max Heap, the parent node is always greater than or equal to its children, making the root the maximum element in the heap. In a Min Heap, the parent node is always less than or equal to its children, making the root the minimum element. Both are complete binary trees used for different priority-based applications.

Correct Answer

 Score : 1 / 1

Evaluation Method : [Embedding](#)

Feedback

Answer is correct

Explanation

The context defines Max Heap and Min Heap based on the relationship between parent and children nodes.

Question 10

Correct

Question

Which of the following is NOT a type of rotation in an AVL Tree?

 Your Answer

RLL Rotation

Correct Answer

RLL Rotation

 Score : 1 / 1

Evaluation Method : [NODE](#)

Explanation

The context lists LL, RR, LR, and RL rotations but not RLL.

Question 11**Incorrect** **Question**

Which data structure is used by Breadth First Search (BFS)?

 Your Answer

Trie

Correct Answer

Queue

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

The context states that BFS uses a Queue.

Question 12

Incorrect

Question

What are the two main collision handling techniques in a Hash Table?

Your Answer

All of the above

Correct Answer

Chaining and Open Addressing

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

The context explicitly mentions Chaining and Open Addressing as collision handling techniques.

Question 13

Incorrect

Question

What is the time complexity of BFS and DFS?

Your Answer

$O(V)$

Correct Answer

$O(V+E)$

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

The context specifies that both BFS and DFS have a time complexity of $O(V+E)$.

Question 14**Correct** **Question**

What is the primary use of a Heap?

 Your Answer

Priority Queue and Heap Sort

Correct Answer

Priority Queue and Heap Sort

 **Score : 1 / 1**

Evaluation Method : [NODE](#)

Explanation

The context specifies that Heaps are used for Priority Queues and Heap Sort.

Question 15**Incorrect** **Question**

Which data structure is primarily used for range queries?

 Your Answer

Binary Search Tree

Correct Answer

Segment Tree

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

Segment Trees are specifically designed for range queries with efficient build, query, and update operations.

Question 16

Incorrect

Question

Which data structure is commonly used in database indexing and file systems?

Your Answer

Heap

Correct Answer

B Tree

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

The context explicitly mentions that B Trees are used in database indexing and file systems.

Question 17

Correct

 Question

What is the balance factor in an AVL Tree?

 Your Answer

Height(left) - Height(right)

Correct Answer

Height(left) - Height(right)

 Score : 1 / 1

Evaluation Method : [NODE](#)

Explanation

The context defines the balance factor in an AVL Tree as Height(left) - Height(right).

Question 18**Correct** **Question**

Which graph representation uses $O(V+E)$ space?

 Your Answer

Adjacency List

Correct Answer

Adjacency List

 Score : 1 / 1

Evaluation Method : [NODE](#)

Explanation

The context specifies that an Adjacency List uses $O(V+E)$ space.

Question 19

Incorrect

Question

What is the time complexity for search, insert, and delete operations in a Red Black Tree?

Your Answer

$O(n \log n)$

Correct Answer

$O(\log n)$

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

The context specifies that all operations in a Red Black Tree take $O(\log n)$ time.

Question 20

Incorrect

Question

What is the time complexity for search, insert, and delete operations in a Hash Table?

Your Answer

$O(n)$

Correct Answer

$O(1)$

 Score : 0 / 1

Evaluation Method : [NODE](#)

Explanation

The context specifies that these operations in a Hash Table take $O(1)$ time on average.

Question 21

Incorrect

Question

Explain the difference between BFS and DFS.

Your Answer

both are travel algorithm

Correct Answer

 Score : 0.2 / 1

Evaluation Method : [LLM](#)

Feedback

The answer is too vague and does not explain the differences between BFS and DFS. Mention key aspects like data structures used, traversal order, and applications.

Explanation

The context provides applications and data structures used by BFS and DFS, which can be used to explain their differences.

Question 22

Incorrect

 Question

What are the key differences between a BST and an AVL Tree?

 Your Answer

The used to tarbael

Correct Answer **Score :** 0 / 1

Evaluation Method : [LLM](#)

 Feedback

The answer is completely unrelated and unclear. The student did not address the differences between a BST and an AVL Tree.

Explanation

The context provides information about BST rules and AVL Tree properties, which can be used to highlight their differences.

Question 23**Correct** **Question**

Why is a Red Black Tree preferred over a regular BST in certain applications?

 Your Answer

Red Black tree is self balcing tree

Correct Answer **Score :** 1 / 1

Evaluation Method : [Embedding](#)

 Feedback

Answer is correct

Explanation

The context mentions that Red Black Trees are self-balancing BSTs with $O(\log n)$ operations, which can be compared to regular BSTs.

Question 24**Incorrect** **Question**

What is the significance of the balance factor in an AVL Tree?

 Your Answer

0,1,-1

Correct Answer **Score :** 0 / 1

Evaluation Method : [LLM](#)

 Feedback

The answer does not explain the significance of the balance factor in an AVL Tree. It only lists the possible values of the balance factor.

Explanation

The context defines the balance factor and mentions that AVL Trees use rotations to maintain balance.

Question 25

Incorrect

Question

What are the advantages of using an Adjacency List over an Adjacency Matrix for graph representation?

Your Answer

this is used to operating System

Correct Answer

 Score : 0 / 1

Evaluation Method : [LLM](#)

Feedback

The answer is completely irrelevant to the question about graph representations. No marks awarded.

Explanation

The context provides space complexities for both representations, which can be used to discuss their advantages.

Question 26

Correct

Question

In a scenario where you need to frequently perform range queries on an array, which data structure would you choose and why?

Your Answer

"I would choose a Segment Tree because it is specifically designed for range queries with efficient build, query, and update operations. The build time is $O(n)$, and both query and

update operations take $O(\log n)$ time, making it suitable for frequent range queries."

Correct Answer

 Score : 1 / 1

Evaluation Method : [Embedding](#)

Feedback

Answer is correct

Explanation

The context highlights the use of Segment Trees for range queries and their time complexities.

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